

Homework 3, Math 609

September 24, 2026

PROBLEM 1:

- **(5 points)** Show that if A is $n \times n$ matrix, then

$$\det(A) = \prod_{i=1}^n \lambda_i,$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A . (Hint: Consider $p_A(0)$, where p_A is the characteristic polynomial of A)

- **(10 points)** Show that if $a_{ii} \neq 0$ for $i = 1, \dots, n$, then

$$\rho(B_w) \geq |w - 1|,$$

where B_w is the iteration matrix of the SOR method. (Hint: Use part 1 of this problem, the fact that $\det(D)^{-1} = \det(D + wL)^{-1}$ and the fact that the determinant of a product of matrices is the product of the determinants of the factors.)

PROBLEM 2: Consider

$$A = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ 5 \\ 4 \end{pmatrix}$$

Observe that the exact solution is $(1, 1, 1)^T$.

- **(5 points)** Find $\|B_J\|_\infty$, $\rho(B_J)$, $\|B_{GS}\|_\infty$, $\rho(B_{GS})$, where B_J and B_{GS} are the Jacobi and Gauss-Seidel iteration matrices, respectively. Use the theorem given in the notes to find the optimal SOR parameter w^* (you should check that A satisfies the hypothesis of the theorem).
- **(10 points)** Perform 10 iterations of Jacobi, Gauss-Seidel and SOR with the optimal parameter w^* obtained in part 1 of this problem, using initial guess $x^{(0)} = 0$. Make a table containing the following information: column 1: k (iteration step number)

column 2: $x_1^{(k)}$ (first component of $x^{(k)}$)

column 3: $x_2^{(k)}$ (second component of $x^{(k)}$)

column 4: $x_3^{(k)}$ (third component of $x^{(k)}$)

column 5: $\|e^{(k)}\|_\infty$ (norm of error at step k)

column 6: $\|e^{(k)}\|_\infty / \|e^{(k-1)}\|_\infty$ (ratio of successive error norms)

Discuss the results.

PROBLEM 3: Consider the basic iteration process given by

$$Qx^{(k+1)} = (Q - A)x^{(k)} + b,$$

where Q is an invertible matrix.

- (5 points) Show that this iteration process is equivalent to the following:

$$x^{(k+1)} = x^{(k)} + z^{(k)},$$

where $z^{(k)}$ satisfies the equation $Qz^{(k)} = r^{(k)}$ with $r^{(k)} = b - Ax^{(k)}$.

- (5 points) Show that

$$r^{(k+1)} = (I - AQ^{-1})r^{(k)}, \quad z^{(k+1)} = (I - Q^{-1}A)z^{(k)}.$$

PROBLEM 4: (10 points) Given the block tridiagonal matrix

$$A = \begin{pmatrix} B & -I & 0 \\ -I & B & -I \\ 0 & -I & B \end{pmatrix}, \quad \text{where} \quad B = \begin{pmatrix} 4 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 4 \end{pmatrix},$$

solve the linear system $Ax = b$ with $b^T = (0, 0, 1, 0, 0, 1, 0, 0, 1)$ using the Jacobi and Gauss-Seidel methods with initial guess $x^{(0)} = 0$. Terminate the iteration when the maximum residual is less than 10^{-2} . To avoid an infinite loop, set the maximum number of iterations to 100. For each method (J,GS) make a table that displays the following information:

column 1: k (number of iterations needed)

column 2: $\|r^{(k)}\|_{\infty}$ (norm of residual at final step)

column 3: $\|r^{(k)}\|_{\infty}/\|r^{(k-1)}\|_{\infty}$ (ratio of successive residual norms at final step)

Summarize the results. Are they consistent with the theorems discussed in the notes?